

Pharmacologic Management of Gout

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Pharmacologic Management of Gout:

Objectives

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1. Describe the mechanism of action, key drug interactions, and key toxicities of medications used for gout.
2. Develop an appropriate pharmacologic plan to treat acute gout.
3. Formulate a rational drug regimen for gout prophylaxis.

Pharmacologic Management of Gout:

Outline

- Acute treatment: Target inflammation
 - NSAIDS*
 - Corticosteroids*
 - Colchicine*
 - Other agents: Uricase, IL-1 inhibitors, ACTH
- Urate-lowering Therapy (ULT): Reduce serum uric acid
 - Xanthine oxidase inhibitors – Allopurinol, febuxostat
 - Uricosuric drugs – Probenecid

Pharmacologic Management of Gout: Objectives

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 - Corticosteroids
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- **Transition to ULT = Anti-inflammatory drug + ULT**
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Pharmacologic Management of Gout: Acute Treatment

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- General principles
 - Treat with pharmacotherapy.
 - Initiate treatment within 24 hours.
 - Ongoing ULT should not be interrupted during an acute episode.

Pharmacologic Management of Gout:

Acute Treatment - NSAIDS

- Mechanism: Inhibit prostaglandin synthesis and urate crystal phagocytosis
- Clinical use: Acute management of gout, transitioning to ULT
- Toxicities: Dyspepsia, GI ulcers, renal toxicity, bleeding
- Naproxen, indomethacin and sulindac are FDA-approved, others are used.
- Full anti-inflammatory doses (vs analgesic doses) should be used.
- **May be poorly tolerated in the elderly, renal insufficiency**

Pharmacologic Management of Gout:

Acute Treatment - Corticosteroids

- Mechanism: Suppression of immune response
- Clinical use: Acute management of gout
- Toxicities: Insomnia, mood changes, hyperglycemia, fluid retention
- Dose:
 - Oral: recommended if 1 or 2 joints involved
 - Prednisone: 0.5 mg/kg/day for 5-10 days at full dose **OR** 2-5 days at full dose then taper for 7-10 days
 - Intra-articular (IA) if 1 or 2 large joints
 - Can be used in combination with oral steroid, colchicine or NSAIDS
 - IA steroids may appear as crystal in joint aspirate
 - Intramuscular (IM)

Pharmacologic Management of Gout:

Acute Treatment - Colchicine

- Mechanism: Inhibits inflammatory response by inhibiting microtubule polymerization & inflammatory mediators
- Clinical use: Acute management of gout, transitioning to ULT; may be used for long-term prophylaxis
- Toxicities: Nausea, vomiting, diarrhea, (neuropathy, bone marrow suppression)
- Avoid in patients with renal or hepatic impairment who are taking P-gp or strong CYP3A4 inhibitors.
 - In these patients life-threatening and fatal colchicine toxicity has been reported with colchicine taken in therapeutic doses.

Pharmacologic Management of Gout:

Acute Treatment - Colchicine

- Recommended if initiated within **36 hours** of symptom onset
- Dose: 1.2 mg (two 0.6 mg tablets) once, then 0.6 mg in 1 hour
 - Total dose is three 0.6 mg tablets. Please know this dose.
 - This dose is lower than that historically recommended but is equally efficacious and less toxic.

Pharmacologic Management of Gout:

Acute Treatment – Other agents

- Biologics
 - Uricase
 - Pegloticase (Krystexxa[®])
- IL-1 Inhibitors
 - Canakinumab (Ilaris[®])
- Adrenocorticotrophic hormone (ACTH)
 - Corticotropin

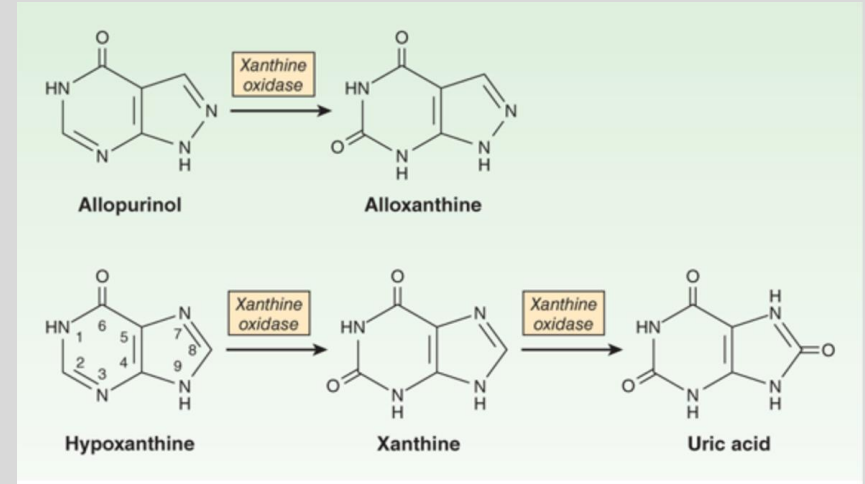
Pharmacologic Management of Gout: Prophylaxis (Urate-Lowering Therapy)

- ULT initiation is recommended if:
 - ≥ 1 tophi
 - Evidence of radiologic damage
 - Frequent attacks (≥ 2 attacks/year)
- ULT may be initiated during an acute attack if an anti-inflammatory medication has been initiated; ULT alone may induce gout flare
- Dose ULT therapy to target a serum uric acid <6 mg/dL.

Pharmacologic Management of Gout:

Prophylaxis – Xanthine Oxidase Inhibitors (XOIs)

- Although most patients under-excrete uric acid (vs over-produce), xanthine oxidase inhibitors are commonly prescribed due to ease of use and efficacy.



Pharmacologic Management of Gout:

Prophylaxis (XOI) - Allopurinol

- Mechanism: Reduces uric acid formation through competitive inhibition of xanthine oxidase
- Clinical use: Chronic management of gout (typically 1st line); also indicated for cancer treatment-related hyperuricemia
- Metabolism: Metabolized by xanthine oxidase to oxypurinol (alloxanthine), which also inhibits xanthine oxidase

Pharmacologic Management of Gout:

Prophylaxis (XOI) - Allopurinol

- Toxicities:
 - Most adverse effects are rare, mild and reversible: dyspepsia, headache, diarrhea
 - If rash develops patient should STOP the medication!!
 - Hypersensitivity is an infrequent complication associated with the HLA-B*5801 allele (rash, eosinophilia, leukocytosis, fever, hepatitis and progressive renal failure)
- Initial dose: 100mg/day
 - Can initiate in CKD $\geq$ stage 3; need to renally adjust dose
- Drug interactions: Azathioprine, mercaptopurine
- Monitoring: Liver function tests (LFTs), serum creatinine (Scr), CBC

Pharmacologic Management of Gout:

Prophylaxis (XOI) - Febuxostat

- Mechanism: Non-purine inhibitor of xanthine oxidase
- Indications: Chronic management of gout
- Metabolism: Hepatic oxidation and conjugation to inactive metabolite; no renal adjustment necessary
- Toxicities: ↑ LFTs, nausea, musculoskeletal pain, rash
- Black box warning (2019): Increased risk of CV death in patients with history of CVD

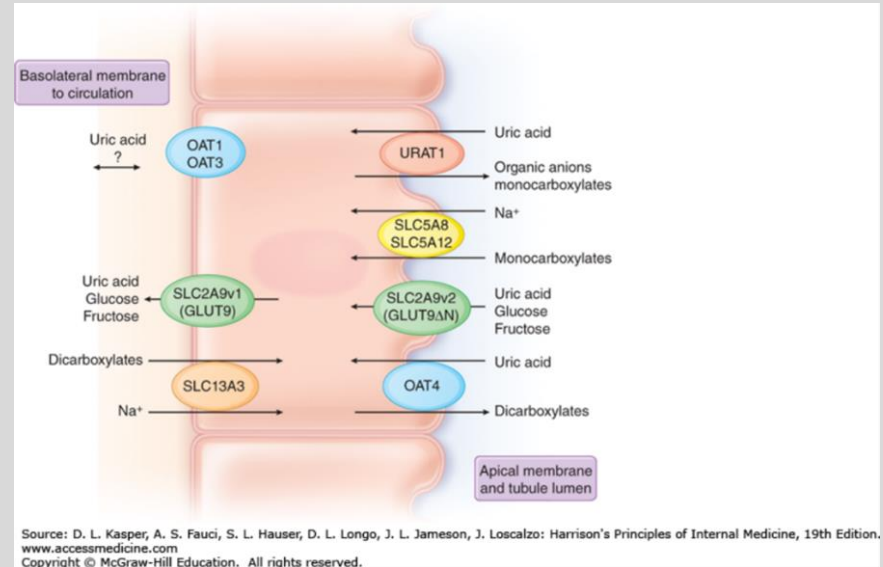
Pharmacologic Management of Gout:

Prophylaxis (XOI) - Febuxostat

- Drug interactions: Azathioprine, mercaptopurine
- Monitoring: LFTs and CBC at baseline and at 2 and 4 months; monitor periodically
- Dose: Initial dose is 40 mg/day

Pharmacologic Management of Gout: Prophylaxis (Uricosuric) - Probenecid

- Uricosurics compete with uric acid for the transporter, inhibiting reabsorption in proximal tubule
- Urine volume must be maintained
- Avoid in:
 - Uric acid over-producers
 - Urolithiasis



Pharmacologic Management of Gout:

Prophylaxis (Uricosuric) - Probenecid

- Mechanism: Competitively inhibits URAT1
- Indication: Chronic management of gout
- Toxicities: Rash, precipitation of gouty arthritis, nephrolithiasis
- Drug interactions: Reduces excretion of weak acids (penicillins)
- Not effective if eGFR <60 mL/min

Duration of Anti-inflammatory Prophylaxis in Combination with ULT

- Transition from acute treatment to prophylaxis:
 - Continue concomitant anti-inflammatory prophylaxis with ULT for 3-6 months.
 - Continuously evaluate for flares.

Pharmacologic Management of Gout:

Summary

- The management of acute episodes should be individualized, based on the patient's co-morbidities.
- If patient taking ULT at onset of an acute attack, ULT should be continued.
- An anti-inflammatory agent should be started before ULT is initiated.
- Core References:
 - Harrison's Principles of Internal Medicine, 21e, 2022; Chapter 372: Gout and Other Crystal-Associated Arthropathies.
 - Katzung's Basic and Clinical Pharmacology, 16e, 2024; Chapter 36. Nonsteroidal Anti-Inflammatory Drugs, Disease-Modifying Antirheumatic Drugs, Nonopioid Analgesics, & Drugs Used in Gout
 - 2020 American College of Rheumatology Guideline for the Management of Gout, Arthritis Care & Research. 2020; 72(6): 744-760.

Lecture and Lab Feedback Form:

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